Mechatronics Engineering and Automation Program
CSE480/CSE482 Lab
Assignment #01



Task 1: Selective Colour Isolation

In this exercise, students will learn how to isolate a specific colour while desaturating the rest of the image. Start by loading the `apple.png` image and creating a grayscale version of it. By splitting the original image into its blue, green, and red channels, you can identify the pixels where the red channel is dominant. Create a binary mask using a threshold on this red channel, then use bitwise operations to 'paste' the original red apple pixels back onto the grayscale background. This demonstrates how masks can selectively combine information from different image versions.

Task 2: Creating a Ghostly Overlay

This task focuses on image blending and weighted arithmetic. Students should load two different images, such as `cameraman.jpg` and `zebra.png`, and resize them so they share the exact same dimensions. Using the `cv2.addWeighted` function, experiment with different transparency levels (alpha and beta values) to make one subject appear like a translucent 'ghost' hovering over the other. This helps in understanding how linear combinations of pixel values affect visual transparency.

Task 3: Artistic Vignette Filtering

Students will create a classic 'vignette' effect that darkens the edges of a photo to draw focus to the center. Load `marsrover.png` and generate a black mask of the same size. Draw a large, white, filled circle in the middle, and apply a significant Gaussian Blur to that mask to create a soft, smooth transition. Finally, combine this mask with your image using multiplication or bitwise AND to see how a smooth-edged mask can create artistic lighting effects.

Task 4: Channel Swapping (Surreal Colours)

This exercise encourages students to rethink how colour images are structured. Load the `zebra.png` image and use `cv2.split` to separate the blue, green, and red channels. Instead of merging them back in their original BGR order, try merging them in a different sequence—for example, putting the green data into the Red channel. Observe how the environment changes into a surreal, alien-like landscape, which illustrates how the computer's interpretation of channel order defines the colours we see.